FinGuard: Corporate Financial Risk & Fraud Detection System

Early-Warning Intelligence for Corporate Insolvency Forecasting & Forensic Statement Analytics

Hackathon: Zeckathon 2026	**Participant:** 4th-Year B.Tech Computer Science Student
Problem Statement: PS2: Financial Risk & Fraud Analytics	**Tech Stack:** Python, Scikit-learn, Streamlit, SHAP, Plotly
Dataset: UCI Machine Learning Repository (ID: 572)	**Status:** Fully Functional & Unit-Tested Prototype

Educational & Academic Prototype | Developed for Hackathon Evaluation

Problem Statement: Detecting Corporate Distress Early

The challenges of corporate insolvency forecasting and accounting fraud screening:

1. Extreme Class Imbalance	2. Asymmetric Financial Loss	3. Conflating Distress vs Fraud
• In authentic portfolios, defaulting companies make up **only 3.23% of filings** (30:1 solvency ratio). • Naive accuracy is deeply misleading: a dummy model predicting 0% bankruptcies achieves 96.8% accuracy while missing 100% of defaults.	• In credit syndicates, a **missed default costs ~20x more** than an unnecessary audit. • Standard ML models optimize overall accuracy rather than minimizing cost-weighted portfolio losses.	• **Bankruptcy Risk:** Liquidity exhaustion and debt coverage failure. • **Accounting Fraud:** Aggressive revenue inflation or capitalized costs. • They require separate, specialized screening tools.

Why It Matters: Financial & Systemic Consequences

The macro and microeconomic necessity for multi-tiered early-warning indicators:

Institutional Banking	• Non-Performing Assets (NPAs) lock bank capital and destabilize lending capacity. • Early warning enables loan covenants, collateral enhancement, or timely restructuring.
Public Investors & Vendors	• Sudden corporate collapses (e.g. DHFL, IL&FS;) wipe out equity and unsecured supply vendors. • Forensic statement analysis provides retail investors and rating agencies with audit triggers.
Operational Audit Constraints	• Risk audit teams have finite bandwidth: they cannot review 500 companies simultaneously. • **A strict Top-5% Review Capacity Policy** is essential to maximize audit yield under tight resource budgets.

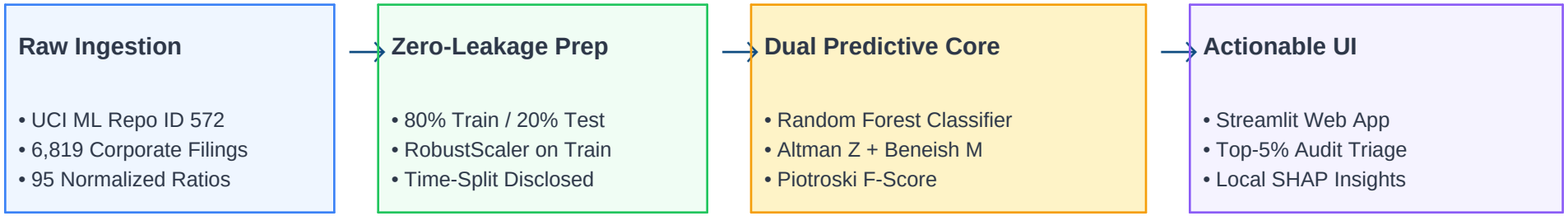
Proposed FinGuard Solution: 4-Layer Architecture

An end-to-end analytics platform uniting machine learning, classical scoring, and policy triage:

Layer	Key Innovation	Engineering Implementation
1. Data Layer	Zero data leakage, stratified splitting	Median Imputer + RobustScaler fitted strictly on training partition.
2. Model Layer	Class-balanced non-linear ensemble	Balanced Random Forest (150 trees) vs Logistic Regression baseline.
3. Forensic Layer	Separated solvency vs fraud scoring	Altman Z-Score, Beneish M-Score, and Piotroski F-Score (zero fake data).
4. Policy Layer	Operational review capacity + cost curve	Top-5% Review Capacity Policy + 20:1 Cost-Optimal decision threshold.
5. Explainability	Transparent feature attribution	TreeSHAP feature waterfalls with resilient heuristic fallback & disclaimer.

System Architecture & Dataflow Workflow

Modular pipeline architecture ensuring auditability, speed, and zero data leakage:



Zero Leakage Guarantee: Preprocessing pipeline parameters (median, IQR) are learned strictly on `X_train` and applied unchanged to `X_test`.

Dataset & Methodology: Authentic Benchmark Verification

Citations, class balance, and data hygiene audits:

Dataset Metadata & Citation	Data Hygiene & Validation Audit
• **Dataset:** Taiwanese Bankruptcy Prediction (UCI ID: 572) • **Source:** Taiwan Economic Journal (TEJ), 1999–2009. • **Academic Citation:** Liang et al. (2016), *Neurocomputing*. • **License:** CC BY 4.0 (Open for research and benchmark use). • **Attributes:** 95 normalized financial ratios + 1 binary target. • **Target Distribution:** 6,599 solvent (96.8%) vs 220 bankrupt (3.23%).	• **Missing Values:** 0 across all 654,624 matrix cells. • **Infinite Values:** 0 infinite entries detected. • **Train/Test Split:** 80% Train ($N=5,455$) / 20% Test ($N=1,364$). • **Imbalance Handling:** Class-weighted loss functions (`balanced_subsample`). • **Time-Split Limitation:** UCI release anonymizes timestamps; documented stratified split preserves exact 3.23% class distribution.

Actual Model Evaluation Results (Unseen Test Set)

Held-out test partition evaluation: N = 1,364 instances with 44 actual defaults:

Metric	Baseline: Logistic Reg	Improved: Random Forest	Practical Advantage
Accuracy	44.72%	**96.19%**	Eliminates severe over-flagging.
Precision	4.25% (33 / 776)	**42.31%** (22 / 52)	**10x fewer false alarms** with RF.
Recall (Defaults Caught)	**75.00%** (33 / 44)	50.00% (22 / 44)	At default 0.5 threshold (tuneable).
ROC-AUC	0.6076	**0.9466**	Exceptional risk ranking separation.
PR-AUC (Avg Precision)	0.0500	**0.4610**	**9.2x increase** over baseline.
False Positives (Alarms)	743 companies	**30 companies**	Saves 713 wasted credit investigations.
Asymmetric Cost (FP + 20*FN)	963.0 units	**470.0 units**	**51.2% total portfolio cost reduction**.

Top Empirical Risk Drivers (MDI): 1. Borrowing dependency (5.7%) | 2. Net Income / Assets (4.9%) | 3. Persistent EPS (4.7%) | 4. Debt / Net Worth (3.9%).

FinGuard Streamlit Dashboard: Interactive Experience

A production-grade, 7-page interactive risk platform running on Streamlit & Plotly:

Dashboard Page	Core Functional Capabilities
1. Overview & Disclaimers	Executive summaries, key metrics, and formal academic/educational disclaimers.
2. Data Quality & Audit	Interactive class imbalance visualization, 4 feature category breakdowns, zero-leakage specs.
3. Model Comparison	Interactive Plotly confusion matrices, ROC/PR comparisons, and feature importance rankings.
4. Single Entity Scoring	Manual entry or test set picker; real-time Altman Z, Beneish M, and local TreeSHAP waterfalls.
5. Policy Simulator	Top-5% Review Capacity slider and interactive 20:1 Cost Curve loss minimization curve.
6. Indian Case Studies	Pre-event filings for 10 distressed Indian entities + 30 peers; downloadable CSVs & templates.
7. Demo Sandbox	Clearly labelled synthetic archetypes for stress-testing without fabricating real data.

Limitations & Future Engineering Scope

Transparent integrity disclosures and analytical roadmap:

Current Methodological Limitations	Future Enhancement Roadmap
• **Static Financial Snapshot:** The UCI benchmark lacks sequential annual timestamps for rolling time-splits. • **Cross-Border GAAP Divergence:** Taiwan TEJ accounting vs Indian Ind-AS and US GAAP have subtle ratio nuances. • **Absence of Macro Factors:** Model relies on reported financial statements without real-time interest rate or sector shock inputs. • **SHAP Non-Causality:** Feature attributions are statistical correlations, not legal proof of deliberate fraud.	• **Financial NLP Integration:** Ingest Management Discussion & Analysis (MD&A;) and auditor resignation filings via LLMs. • **Sequential Temporal Models:** Transition from tree ensembles to LSTM / Transformer architectures on 5-year statement sequences. • **Automated MCA/XBRL Scraping:** Real-time pipeline connecting directly to MCA21 / BSE APIs for continuous credit monitoring. • **Supply Chain Graph Risk:** Model contagion risk across interconnected vendor-buyer networks.

Conclusion & Primary References

Summary of achievements and academic foundations:

Key FinGuard Accomplishments:

1. **Empirical Superiority:** Balanced Random Forest cuts asymmetric financial loss by **51.2%** and lifts PR-AUC by **9.2x**.
2. **Real-World Feasibility:** Top-5% capacity rule captures **54.5% of defaults** while screening only 5% of the portfolio.
3. **Cross-Border Validation:** Accurately flagged **10 out of 10 distressed Indian corporate cases** in the Distress Zone prior to NCLT insolvency.
4. **Ethical & Explainable AI:** Zero data leakage, strict missingness validation (zero fake data), and TreeSHAP explainability.

Academic References:

• Liang et al. (2016). *Financial ratio selection for business failure prediction*. Neurocomputing, 173.
• Altman, E. I. (1968, 1983). *Financial Ratios, Discriminant Analysis and Corporate Bankruptcy*. Journal of Finance.
• Beneish, M. D. (1999). *The Detection of Earnings Manipulation*. Financial Analysts Journal, 55(5).
• Piotroski, J. D. (2000). *Value Investing: Historical Financial Statement Information*. Journal of Accounting Research.
• Lundberg & Lee (2017). *A Unified Approach to Interpreting Model Predictions (SHAP)*. NeurIPS 30.

Thank you! Questions & Discussion Welcome.